

orbit_demo simulation report

Package section 05_simulation of orbit_demo_design_review, written 2026-10-06. Links are relative to this file. Paths in backticks without a link (reports/..., build/..., scripts/..., scratch paths) are repository, build or scratch paths that are not copied into the package. Each result carries one status label: VERIFIED (a tool log or report shows it), REPRODUCED 2026-10-06 (re-run for this package), TARGET, ASSUMPTION, CLAIM (unverified), MISSING or N/A. Formal verification, multi-corner STA and physical verification are in [07_verification](#). Discrepancies are collected in the discrepancy register ([09_review_notes](#)).

1 Design revision covered by this package

Package revision table, verbatim from the package revision record:

Item	Revision / identifier	Note
Design	orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)	top module orbit_demo, LANES=4
RTL	rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26566f0 (2026-09-29 18:54 UTC); unchanged through the package commit	sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256 at the package root)
Specification	docs/SPEC.md at 26566f0 (never revised)	sha256 aef5d9fb...
Concept brief	docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)	sha256 9d5f33d3...
Reviewed layout	ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/	6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spef 014e655a...
Layout configuration	pd/sky130hd_sep/config.mk and regions.tcl at c10c59b; constraint.sdc at 21917cf (set clk_period 7.2)	The run (21:56 UTC) used config.mk/regions.tcl as in ed553f8, identical to c10c59b once comments are stripped. It used the working-tree constraint.sdc with period 7.2 ns, which was committed in 21917cf at 22:00 UTC and is byte-identical to the flow copy build/pd_sep/sdc/sky130hd/sep.sdc. ed553f8's constraint.sdc has a 7.0 ns period, so a rebuild must not use it
Flow / tools	ORFS docker image tag openroad/orfs:latest; the only local image is sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC, before the run). Linking the run to this digest is an ASSUMPTION: the run logged no digest. OpenROAD prints version "unknown". KLayout 0.30.12 (DRC/LVS). Yosys 0.69+154 (local sims/schematics)	No PDK commit is recorded in the logs
Process / library	SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)	—
Package assembled	2026-10-06; design inputs copied from the repository tree at commit 0495cfa (branch claud/hopeful-rubin-0yf8io)	Every later commit on the branch changes only review/ or is empty (git diff --name-only 0495cfa..HEAD outside review/ is empty). Re-runs made for this package are dated 2026-10-06

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout.

Note on the table (checked 2026-10-06): MANIFEST.sha256 is at the package root ([../MANIFEST.sha256](#), generated last on 2026-10-06; discrepancy register D-53). The full sha256 of the four RTL files are also listed in [04_digital_design/README.md](#), notes on the revision table.

1.1 Simulation inputs and tools

Item	Value	Source	Status
RTL simulated	\$(RTL) = rtl/orbit_keep_reg.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_demo.v. rtl/ecc/ is not part of orbit_demo and is not simulated in this section	make_sim-directed.log line 1; RTL defined in Makefile lines 15-17	VERIFIED

Item	Value	Source	Status
RTL at the waveform run	md5 dc59d4a5... (orbit_keep_reg.v), b34f737f... (orbit_mac_lane.v), b2877bc2... (orbit_thermal_tmr.v), 998888fd... (orbit_demo.v), repo HEAD 0495cfa6; equal to the current rtl/ (sha256 as in the revision table, checked 2026-10-06)	waveforms/run_info.txt	REPRODUCED 2026-10-06
Directed bench	tb/tb_orbit_demo.v, sha256 2d7a33f4830039a7..., last changed in commit f057f38 (2026-09-30 00:31 UTC), the same commit that last changed reports/sim/	tb_orbit_demo.v ; <code>git log -- tb/tb_orbit_demo.v reports/sim</code>	VERIFIED
Published logs	published_logs/ are byte-identical copies of reports/sim/* (cmp, 2026-10-06), "Generated by make sim-report on 2026-09-30" (summary.md line 3), file times 2026-09-30 00:25 UTC. They carry no commit id, RTL hash or date	summary.md	VERIFIED
Re-run logs	rerun_logs_2026-10-06/ : the same make targets at repo commit 0495cfa, 2026-10-06. The four RTL files of \$(RTL) (26566f0), tb/tb_orbit_demo.v, tb/cocotb/, tb/sim_mutants.py and tb/sim_report.py (f057f38), and mk/sim.mk (3196533, 2026-09-29 20:56 UTC) are unchanged since the published run (rtl/ecc/ and tb/ecc/ changed later in e80a7b2 but are not used by the sim targets). review_runs/ holds the review logs of make sim, make sim-mutants, make formal-quick and make model MODEL_PUBLISH= quoted in 04_digital_design section 6.3	make_*.log in that folder	REPRODUCED 2026-10-06
Icarus Verilog, re-runs	14.0 (devel) (s20260301-500-g2e81fccb-dirty)	run_info.txt line 3; requirements-tools.txt line 8	VERIFIED
Icarus Verilog, 2026-09-30 runs	same build as the re-runs (same /opt/eda/oss-cad-suite install); the published directed and GLS logs do not print the Icarus version	—	ASSUMPTION
Verilator	5.053 devel rev v5.052-233-gf5f9ddef9 (mod)	directed_verilator.log line 29; make_lint.log line 2	VERIFIED
Yosys	0.69+154 (git sha1 30d62572e-dirty)	gls_yosys_yosys.log line 1538	VERIFIED
cocotb / Python	cocotb 2.1.0.dev0+41564633, Python 3.11.6 (/opt/eda/oss-cad-suite/bin/tabbypy3)	random_icarus.md line 3	VERIFIED
python3 for the model self-test	not printed in any log	—	MISSING
Published tool line	summary.md line 19 lists every tool version as "unknown" although the copied logs state them; see discrepancy register (09_review_notes)	summary.md line 19	VERIFIED

2 Test inventory

Common conditions unless stated: functional simulation, zero delay, 10 ns simulation clock (HALF = 5, timescale 1ns/1ps). The 10 ns period has no timing meaning; the reviewed layout clock is 7.2 ns ([clock_period.txt](#)) and no simulation runs at it. No PVT corner applies (section 5). Formal property checking is not repeated here: see [07_verification/formal_rerun_2026-10-06/formal_summary.md](#).

Table 2a: tests, stimulus and tools.

#	Test	What it checks	Stimulus / size	Simulator + version
S1	Directed bench, RTL, Icarus (make sim-directed)	tb/tb_orbit_demo.v, ports only: cycle-accurate SPEC model compared every cycle on in_ready, out_valid, out_data, therm_state, shutdown_req, fault (= 0), therm_repair (= 0); scoreboard on every out_fire; scenario checks with hand-computed literals	14 scenarios, +seed=1 (in-bench xorshift32), 133601 cycles, 133209 beats, 401 results	Icarus 14.0 (devel), iverilog -g2005 -Wall -Wno-timescale, vvp -n

#	Test	What it checks	Stimulus / size	Simulator + version
S2	Directed bench, RTL, Verilator (make sim-verilator)	S1 bench in a 2-state simulator; summary block must equal S1	as S1	Verilator 5.053 devel, --binary --timing, -O0
S3	cocotb constrained random, Icarus (make sim-random)	all 7 outputs vs Python model tb/cocotb/orbit_ref.py every cycle; scoreboard on every out_fire; 35 required coverage points (random_workload), 5 (wrap_soak)	random_workload seeds 1-5 x 20000 cycles; wrap_soak seed 1, 149226 cycles (500 cycles after both INT32 wrap directions, limit 400000); 249226 cycles total	Icarus 14.0 (devel), cocotb 2.1.0.dev0+41564633, Python 3.11.6
S4	cocotb random, Verilator (make sim-random-verilator)	as S3; every coverage count must equal the Icarus run of the same seed	random_workload seeds 1, 2 x 20000 cycles; no wrap_soak	Verilator 5.053 devel, cocotb as S3
S5	Reference model self-test (make sim-model)	orbit_ref.py against hand-written cases (products, thermal table incl. code 3, reset/recovery, alternate admission, wraparound, backpressure, clear_fault)	38 checks	python3 (version not logged); no HDL simulator
S6	Directed bench on Yosys generic netlist (make sim-gls)	S1 bench on synth - flatten -top orbit_demo netlist; summary must equal S1	as S1; netlist 4899 cells, 521 flip-flop cells (515 \$ _SDFFE_PN0P_, 3 \$ _SDFFE_PN1P_, 2 \$ _SDFFE_PP0P_, 1 \$ _SDFF_PN0_)	Yosys 0.69+154; Icarus 14.0 (devel); zero delay
S7	Directed bench on routed sky130hd BASE netlist (published GLS; not the reviewed layout)	S1 bench on netlist N1 (base layout, 7.0 ns reference run); summary must equal S1	as S1	Icarus (version not logged); cell models C1; zero delay, no SDF
S8	Directed bench on routed sky130hd SEP netlist (reviewed layout; new)	S1 bench on netlist N2 (reviewed sep layout); summary must equal S1	as S1	Icarus 14.0 (devel); cell models C1 (identical to C2); zero delay, no SDF; compile.log empty
S9	GLS fault campaign on the SEP netlist (make pd-sep-gls)	pd/gls/tb_gls.v: RTL and netlist N2 in lockstep, all outputs compared every cycle; one redundant bit flipped in both models every 32 cycles at the falling edge; duplicated-copy flip must give out_valid = in_ready = 0 that cycle and fault = 1 after the next edge; thermal-copy flip must give therm_repair = 1 with the voted state unchanged, then 0	30000 cycles, seed 1, run 2026-09-29 22:12-22:14 UTC; 936 flips (702 duplicated-copy + 234 thermal-copy) over 518 redundant bits (512 + 6)	Icarus (iverilog -g2005, version not logged); cell models C2; 10 ns clock (tb_gls.v line 161); zero delay
S10	Sim negative control (make sim-mutants, tb/sim_mutants.py)	each of 26 single-bug RTL copies must fail both benches	26 mutants; directed +maxerr=1; random 1 seed x 20000 cycles	Icarus; cocotb as S3

#	Test	What it checks	Stimulus / size	Simulator + version
S11	Mutation campaign (make mutation, scripts/mutation_test.py)	40 single-point RTL mutants run through lint, sim, formal-quick, fault-quick, synth	m01-m40 + baseline m00, 2026-09-30 01:22-02:27 UTC; MUTATION_JOBS=2, MUTATION_TIMEOUT=3600 s (mk/mutation.mk lines 29-32)	per the m00 baseline logs (build/mutation/m00/logs/): Verilator 5.053 devel (lint, sim), cocotb 2.1.0.dev0+41564633 / Python 3.11.6 (sim), Yosys 0.69+154 (synth); Icarus (sim, fault-quick) and SymbiYosys (formal-quick) versions not logged
S12	Waveform scenarios (new, waveforms/tb_orbit_wave.v)	5 self-checking scenarios with VCD dump (section 4)	11 / 21 / 19 / 97 / 24 cycles; 29 / 35 / 42 / 383 / 50 checks	Icarus 14.0 (devel) s20260301-500-g2e81fccb-dirty; RTL; zero delay
S13	Waveform negative control (waveforms/run_negctl.sh)	each scenario must FAIL on an RTL copy with its targeted bug (section 6)	6 mutants x 5 scenarios	as S12
S14	Lint (make lint; static, listed because its log is in this folder)	Verilator --lint-only -Wall on the 4 RTL files, top orbit_demo	—	Verilator 5.053 devel rev v5.052-233-gf5f9ddef9 (mod)

Table 2b: pass criteria, results, status and logs (same row numbers).

#	Pass criterion	Result	Status	Log in the package
S1	errors = 0, model checks > 0, out_fire > 0 → TB_ORBIT_DEMO PASS (tb lines 976-982); make greps that line	PASS: 133600 model checks (0 mismatches), 107 scenario checks (0 failed), 152 literal lane checks, 0 errors	REPRODUCED 2026-10-06	published ; re-run
S2	PASS and summary diff vs S1 empty (mk/sim.mk)	PASS; "sim-verilator: summary identical to Icarus"	REPRODUCED 2026-10-06	published ; re-run ; clean build
S3	simulator exit 0, one passing xunit testcase, errors = 0, no required coverage point at 0 (run_random.py lines 30-64)	SIM_RANDOM PASS: 6 runs, 249226 cycles, 7544 results checked, 0 mismatches	REPRODUCED 2026-10-06	published ; re-run summary ; re-run log
S4	as S3, plus "coverage identical" per seed	SIM_RANDOM PASS: 2 runs, 40000 cycles, 1844 results checked, 0 mismatches; coverage identical (summary.md lines 137-138)	REPRODUCED 2026-10-06	published ; re-run
S5	prints PASS	orbit_ref self-test PASS (38 checks)	REPRODUCED 2026-10-06	published ; re-run
S6	PASS and summary identical	PASS; "sim-gls: summary identical to the RTL run"	REPRODUCED 2026-10-06	published ; re-run ; Yosys log lines 1406-1418
S7	PASS and summary identical	PASS	VERIFIED	gls_sky130hd.log ; netlist named in summary.md lines 16, 59 and in the file table of build/sim/gls/sky130hd/tb.vvp
S8	PASS and summary identical	PASS; "sim-gls: summary identical to the RTL run"; new run, no committed counterpart	REPRODUCED 2026-10-06	gls_sky130hd_sep_directed.log ; make log
S9	0 mismatches and all stop/repair checks → GLS PASS	GLS PASS: 30000 cycles, 0 mismatches; 702 stopped, 234 repaired, 518 of 518 bits hit	REPRODUCED 2026-10-06	published 07_verification/gls/gls.log (= reports/pdsep/gls.log = build/pd_sep/gls/sep/gls.log); re-run

#	Pass criterion	Result	Status	Log in the package
S10	kill = TB_ORBIT_DEMO FAIL (directed), >= 1 model mismatch (random); all 26 killed by each bench	"directed bench killed 26/26, random bench killed 26/26"	VERIFIED	mutants.md line 34; summary.md line 17
S11	mutant killed if a target fails; baseline passes all	35/40 killed (87.5 %); survivors m07, m20, m21, m22, m27; sim target kills 26	VERIFIED	summary.md , results.json (byte-identical copies of reports/mutation/); see discrepancy register (09_review_notes)
S12	WAVE_<name> PASS (0 failed checks)	5 of 5 PASS	REPRODUCED 2026-10-06	run_scen1.log , run_scen2.log , run_scen3.log , run_scen4.log , run_scen5.log
S13	target scenario FAIL	6 of 6 caught	REPRODUCED 2026-10-06	negctl_summary.txt
S14	no %Warning or %Error line	none printed; "Built from 0.091 MB sources in 6 modules"	REPRODUCED 2026-10-06	make_lint.log

Netlists and cell models used by the gate-level runs (sha256 computed 2026-10-06):

Id	File	sha256 (prefix)	In the package	Status
N1	build/pd/results/sky130hd/orbit_demo/base/6_final.v (base layout, written 2026-09-29 20:10 per summary.md line 59)	b2a28a7b19f16cab	no	VERIFIED
N2	build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.v (reviewed sep layout); S9 simulates build/pd_sep/gls/sep/orbit_demo_gl.v, N2 with the top renamed by pd/gls/gen_gls.py	68093c41fca50e9e	6_final.v.gz decompresses to the same sha256	VERIFIED
C1	build/pd/gls/sky130hd/cells.v	1d60965c2a6e3134	no	VERIFIED
C2	build/pd_sep/gls/sep/cells.v	1d60965c2a6e3134 (byte-identical to C1)	no	VERIFIED

C1 and C2 are Yosys 0.69+154 `read_liberty -ignore_miss_func ...; write_verilog` output of `sky130_fd_sc_hd__tt_025C_1v80.lib` (nom_process 1.0, nom_temperature 25, nom_voltage 1.8). Only the Boolean cell functions are used; there are no delays, timing checks or SDF.

Notes:

- S7 vs S8: the published routed-netlist GLS (S7) used the BASE layout netlist. The bench output does not name the netlist; S7, S8, S6 and S1 logs are byte-identical (md5 05a30e4e...). The generator `tb/sim_report.py` hardcodes the base path in [summary.md](#); see discrepancy register (09_review_notes). S8 was run for this package on 2026-10-06 19:35:58-19:37:55 UTC (build file times) and is the only directed-bench GLS of the reviewed netlist in the package.
- S9 compares the RTL with the netlist under the same injection; it has no golden-result scoreboard. The claim that no wrong result is released rests on the stop checks plus the formal property `D_out_fire_matches_ref` (07_verification). Thermal code 3 is never reached (code3 0).
- `make synth` also runs the directed bench on its own `-noexpr` Yosys netlist (`reports/synth/gls_result.txt` = [07_verification/synth/gls_result.txt](#)); it is not reviewed here.

3 Result details

3.1 Directed bench (S1, S2, S6, S7, S8)

Summary block printed identically by all five runs ([directed_icarus.log](#) lines 16-27):

```
tb_orbit_demo summary (seed 1)
  scenarios          14
  cycles             133601
  model cycle checks 133600 (mismatches 0)
  scenario checks    107 (failed 0)
  beats accepted     133209
  results transferred 401 (all compared with the scoreboard)
  literal lane checks 152
```

```

back-to-back results 220
results dropped by clear_fault/reset 2
errors 0
TB_ORBIT_DEMO PASS

```

Scenarios and start times (ps, same log lines 1-14): reset_stop 10000, signed_edges 260000, multi_beat 440000, first_and_last 6830000, continue_after_last 6890000, backpressure 6960000, back_to_back 11250000, throttle 11910000, stop_and_recover 13610000, drain_in_stop 14040000, negative_temps 14210000, int32_wrap 14580000 (132106-beat sum, INT32 wrap in both directions), clear_fault 1335650000, reset_midrun 1335870000; \$finish at 1336020000 ps.

Scope limits stated in [summary.md](#) lines 187-192 (VERIFIED as text): no upsets are injected, so fault and therm_repair are only checked to stay 0 and the mismatch/fault-stop path is not exercised; thermal code 3 is not reachable through the ports; gate-level runs are zero delay; coverage is testbench event counting only.

3.2 cocotb random (S3, S4)

Test	Seed	Cycles	Results checked	Errors	Missing coverage	Icarus	Verilator
random_workload	1	20000	866	0	none	PASS	PASS
random_workload	2	20000	978	0	none	PASS	PASS
random_workload	3	20000	976	0	none	PASS	not run
random_workload	4	20000	825	0	none	PASS	not run
random_workload	5	20000	877	0	none	PASS	not run
wrap_soak	1	149226	3022	0	none (wrap_pos 2, wrap_neg 2)	PASS	not run

Source: [random_icarus.md](#) lines 5-12 and coverage table lines 14-59; [random_verilator.md](#) lines 5-8.

Conditions: cocotb Clock(dut.clk, 10, unit="ns"), inputs written after the rising edge, outputs sampled at the falling edge; seeds and cycle counts from [mk/sim.mk](#) lines 35-40. Status: REPRODUCED 2026-10-06.

reset_cycles per random_workload seed is 6 / 2 / 4 / 7 / 5, of which 2 are the initial reset, so seed 2 has no mid-run reset; reset_cycles is not a required coverage point.

3.3 GLS fault campaign (S9)

[07_verification/gls/gls.log](#), lines 1-6, VERIFIED; lines 1-6 of the 2026-10-06 re-run [gls_rerun.log](#) are identical (REPRODUCED 2026-10-06):

```

GLS: 30000 cycles, seed 1, 518 redundant bits, injection every 32 cycles
coverage: beats accepted 8729, results taken 1037 (1037 non-zero), resets 6, clears 733
coverage: cycles NORMAL 14075, THROTTLE 1737, STOP 14182, code3 0; throttle-blocked beats 525
injections: 702 duplicated-copy flips (702 stopped), 234 thermal-copy flips (234 repaired); 518 of 518 redundant bits hit
GLS PASS: 30000 cycles, 0 mismatches
pd/gls/tb_gls.v:262: $finish called at 300004000 (1ps)

```

The 518 targets are listed in [gl_inject.vh](#) (N_DUP_BITS = 512, N_TMR_BITS = 6), e.g.

rtl.g_lane[0].u_lane.u_acc_a.q[0] paired with gl.\g_lane[0].u_lane.u_acc_a/q[0]\$_SDFFE_PN0P_.IQ. The flow is [mk/pdsep.mk](#) lines 127-137 (PDSEP_GLS_CYCLES ?= 30000, PDSEP_GLS_SEED ?= 1, lines 35-36). The base netlist has its own campaign with a byte-identical log (reports/pd/sky130hd/gls.log), because stimulus and injection schedule depend only on seed and cycle count.

3.4 Sim negative control (S10)

All 26 mutants killed by both benches ([mutants.md](#)). Weakest kills (most cycles needed, random bench):

reset_keeps_out_valid "6 mismatches in 20000 cycles" (line 24), no_drain_in_stop 11804 cycles (line 29), stop_gt 4762 (line 9), reset_normal 3952 (line 14). Directed-bench first errors range from t=24000 ps to t=1335744000 ps. Status: VERIFIED. A scratch re-run on 2026-10-06 produced a byte-identical mutants table; its log is [make_mutants.log](#) (table lines 4-37, SIM_MUTANTS PASS line 39), copied into the package on 2026-10-06. Status of the re-run: REPRODUCED 2026-10-06.

3.5 Mutation campaign (S11)

[summary.md](#) (= reports/mutation/summary.md): score 35/40 (87.5 %), 35/39 (89.7 %) excluding m21 argued equivalent; kills per target lint 5, sim 26, formal-quick 29, fault-quick 21, synth 29. Survivors m07

result_from_b, m20 out_valid_no_fault, m22 fault_not_sticky, m27 shutdown_only_stop are called real test gaps; m21 out_fire_unmasked an equivalent mutant. Status: VERIFIED for the report content; the logged run

ended with unanalysed survivors and the summary was regenerated by a run with no log, see discrepancy register (09_review_notes). This campaign and S10 both report "26" sim kills with different mutant sets.

4 Waveforms

All waveforms come from the wrapper [tb_orbit_wave.v](#), written for this package (tb/ is not modified). Its DUT instance dut has the same named port connections as tb/tb_orbit_demo.v and no parameter overrides, so the internal paths are the SPEC section 7 paths (dut.g_lane[i].u_lane.u_acc_b.q, dut.u_thermal.u_copy1.q). Conditions for every figure: RTL, zero delay, Icarus 14.0 (devel) (s20260301-500-g2e81fccb-dirty), iverilog -g2005 -Wall -Wno-timescale (empty [compile.log](#)), **10 ns functional clock** (rising edges at 5, 15, 25 ns ...), inputs and upsets applied at the falling edge, outputs sampled 1 ns before the rising edge. This is not the 7.2 ns layout clock; the plots show function only. Upsets are hierarchical assignments from the testbench; the RTL has no fault port. Run 2026-10-06T19:46:47Z ([run_info.txt](#)), scripts [run_waves.sh](#) and [plot_waves.py](#) (every plotted value is read from the VCD). Per-cycle values are in the CSV next to each PNG; all six pages are in [orbit_demo_waveforms.pdf](#). Status of every figure: REPRODUCED 2026-10-06.

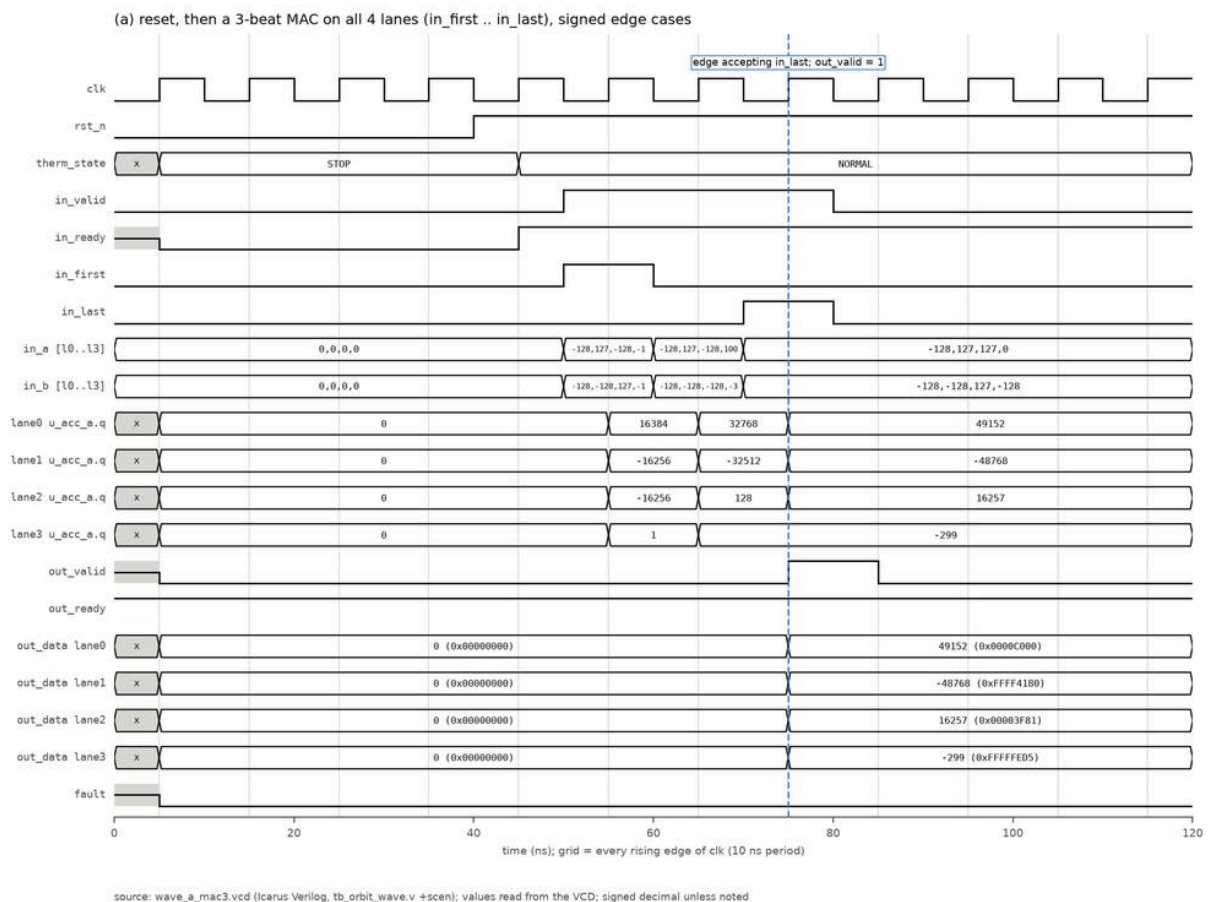


Figure 1, scenario 1 a_mac3. Reset, STOP → NORMAL at 60 C, then a 3-beat sum on all 4 lanes (in_first on beat 1, in_last on beat 3). Lane results: $(-128 \times -128) \times 3 = 49152$ (0x0000C000); $(127 \times -128) \times 3 = -48768$ (0xFFFF4180); $-128 \times 127 + -128 \times -128 + 127 \times 127 = 16257$ (0x00003F81); $-1 \times -1 + 100 \times -3 + 0 \times -128 = -299$ (0xFFFFFED5). out_valid is set by the edge that accepts in_last (75 ns) and the result transfers at the next edge (out_ready = 1). The bench also checks u_acc_b.q = u_acc_a.q after every beat (not plotted) and fault = therm_repair = mismatch = 0. Self-check: WAVE_a_mac3 PASS (29 checks, 11 cycles). Source: [wave_a_mac3.vcd](#), [run_scen1.log](#), [wave_a_mac3.csv](#).

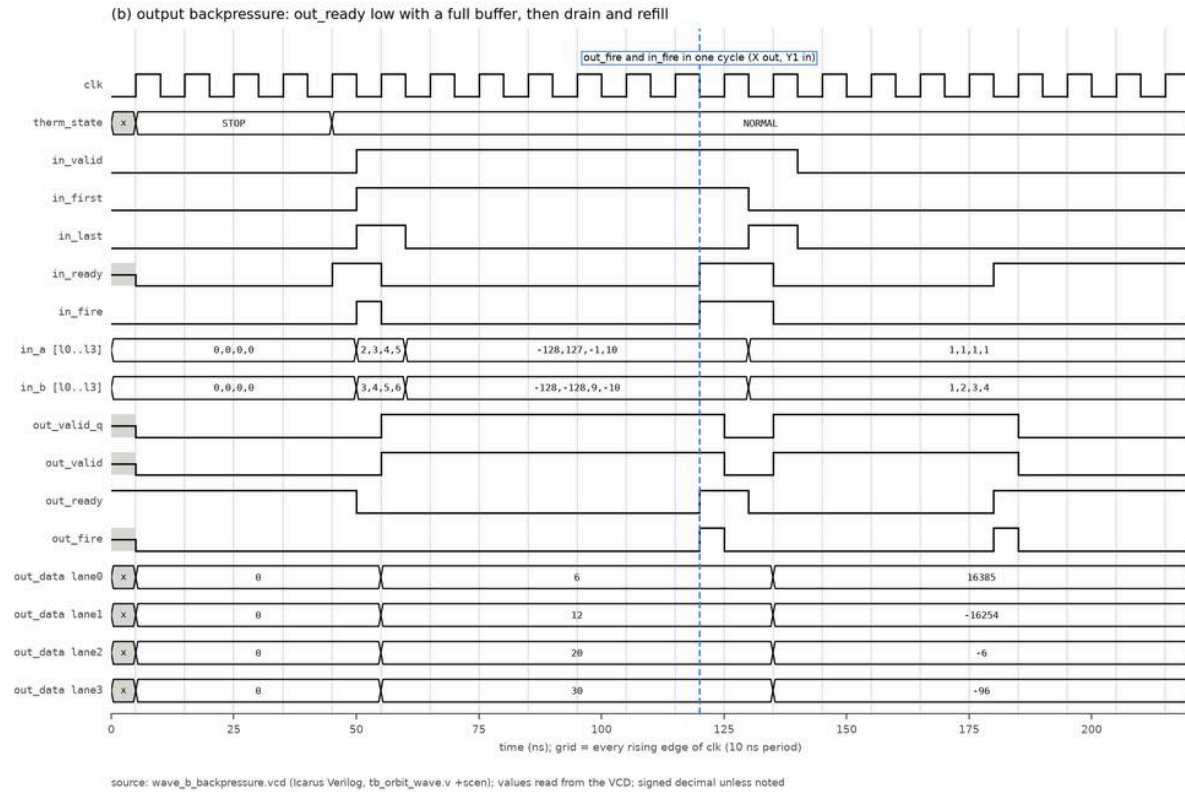


Figure 2, scenario 2 b_backpressure. Result X = (6, 12, 20, 30) is held 6 cycles with out_ready = 0: out_valid = 1, out_data stable, in_ready = 0, and the offered non-last beat Y1 (in_first) is blocked. In the cycle ending at the 125 ns edge X transfers and Y1 is accepted (out_fire and in_fire in the same cycle). Y2 (in_last) enters the empty buffer; Y = (16385, -16254, -6, -96) is held 4 cycles, then transferred. 3 beats accepted, exactly 2 results. Self-check: WAVE_b_backpressure PASS (35 checks, 21 cycles). Source: [wave_b_backpressure.vcd](#), [run_scen2.log](#), [wave_b_backpressure.csv](#).

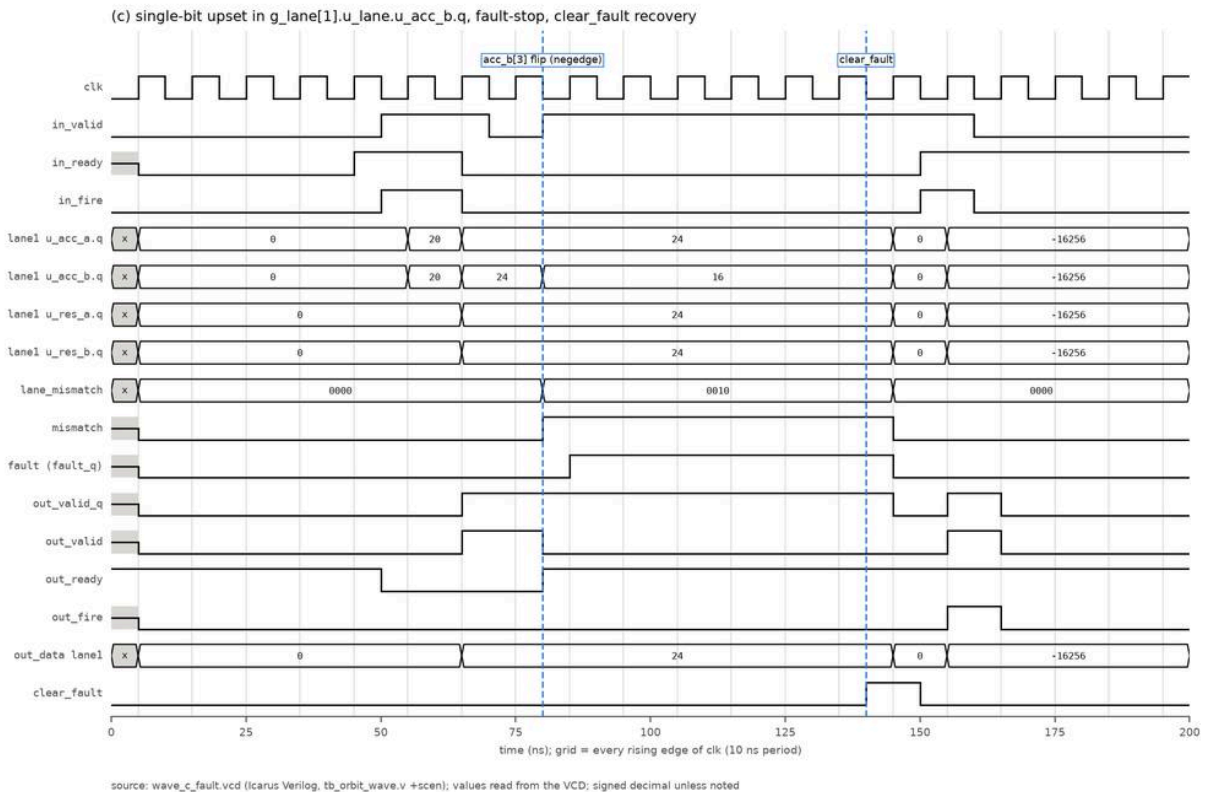


Figure 3, scenario 3 *c_fault*. With result $R = (12, 24, 36, 48)$ pending and $out_ready = 0$, $out.g_lane[1].u_lane.u_acc_b.q[3]$ is flipped at the 80 ns falling edge (lane 1 acc_b 24 $\rightarrow$ 16). In the same cycle $lane_mismatch = 0010$, $mismatch = 1$ and $out_valid = in_ready = 0$; fault ($fault_q$) = 1 after the 85 ns rising edge. For the next 5 cycles, with $out_ready = 1$ and a beat offered, there is no out_fire and no in_fire ; out_valid_q stays 1 internally and R is never transferred. One $clear_fault$ cycle (140-150 ns, $in_ready = 0$ during it) zeroes all acc/res copies, clears fault and mismatch and empties the buffer; the recovery beat (lane 1: $127 \times -128 = -16256$) transfers. Self-check: *WAVE_c_fault* PASS (42 checks, 19 cycles). Source: [wave_c_fault.vcd](#), [run_scen3.log](#), [wave_c_fault.csv](#).

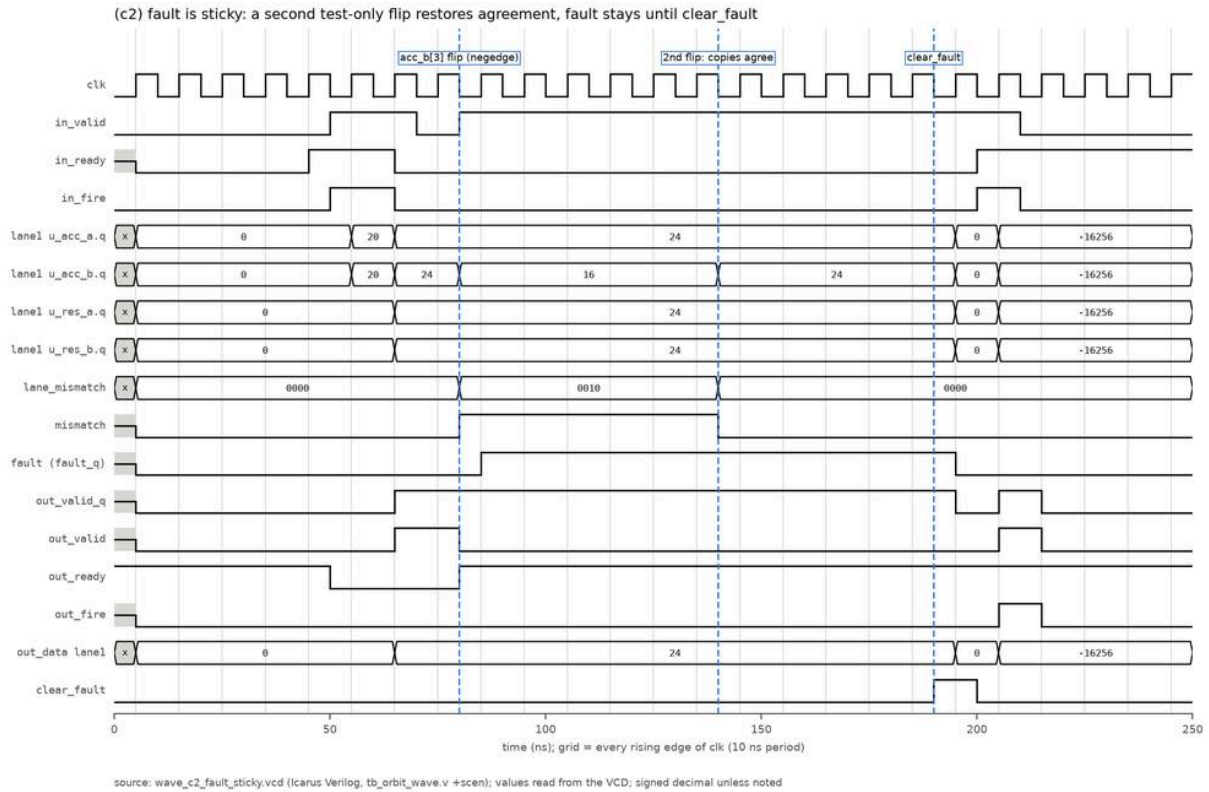


Figure 4, scenario 5 *c2_fault_sticky*. As scenario 3, plus a second, test-only flip of the same bit at 140 ns that restores $acc_b = 24$. $lane_mismatch$ returns to 0000 and $mismatch$ to 0, but fault stays 1 and out_valid and in_ready stay 0 until $clear_fault$ (190-200 ns): the fault latch is sticky (SPEC section 5). Without this scenario a non-sticky fault latch passes scenario 3, because the persistent flip keeps mismatch high (section 6). Self-check: *WAVE_c2_fault_sticky* PASS (50 checks, 24 cycles). Source: [wave_c2_fault_sticky.vcd](#), [run_scen5.log](#), [wave_c2_fault_sticky.csv](#).

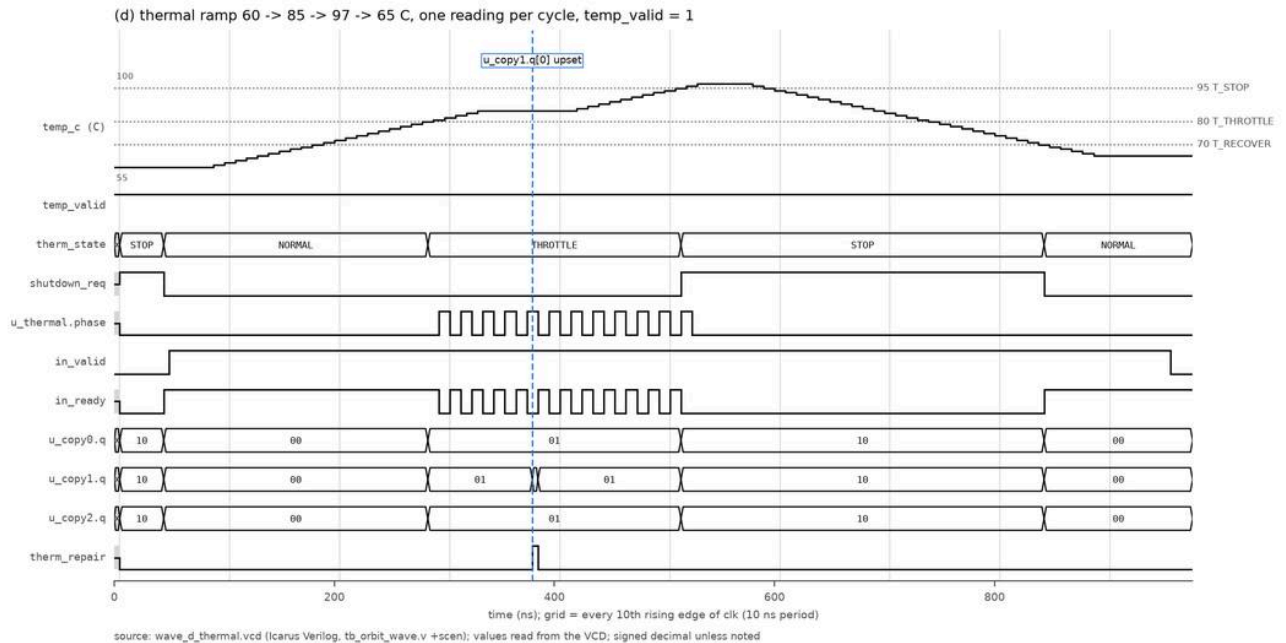


Figure 5, scenario 4 d_thermal. temp_c one reading per cycle: 60 x4, 61..85, 85 x8, 86..97, 97 x4, 96..65, 65 x6 (91 readings), temp_valid = 1, single-beat sums offered every cycle, out_ready = 1. therm_state and in_ready match the SPEC section 6 table at every step. THROTTLE starts at ramp step 24 (the cycle after the 80 C reading), STOP at step 47 (after 95 C), NORMAL at step 80 (after 70 C on the way down); STOP never falls back to THROTTLE. 23 THROTTLE cycles admit 12 (alternate cycles, the first THROTTLE cycle admits). shutdown_req = therm_state[1]. 47 beats accepted, 47 results transferred. In the first STOP cycle u_thermal.phase is still 1 (run_scen4.log step 47); the RTL clears it at the next edge, which differs from the SPEC wording but has no effect on admission; see discrepancy register (09_review_notes). Self-check: WAVE_d_thermal PASS (383 checks, 97 cycles). Source: [wave_d_thermal.vcd](#), [run_scen4.log](#) (per-step table), [wave_d_thermal.csv](#).

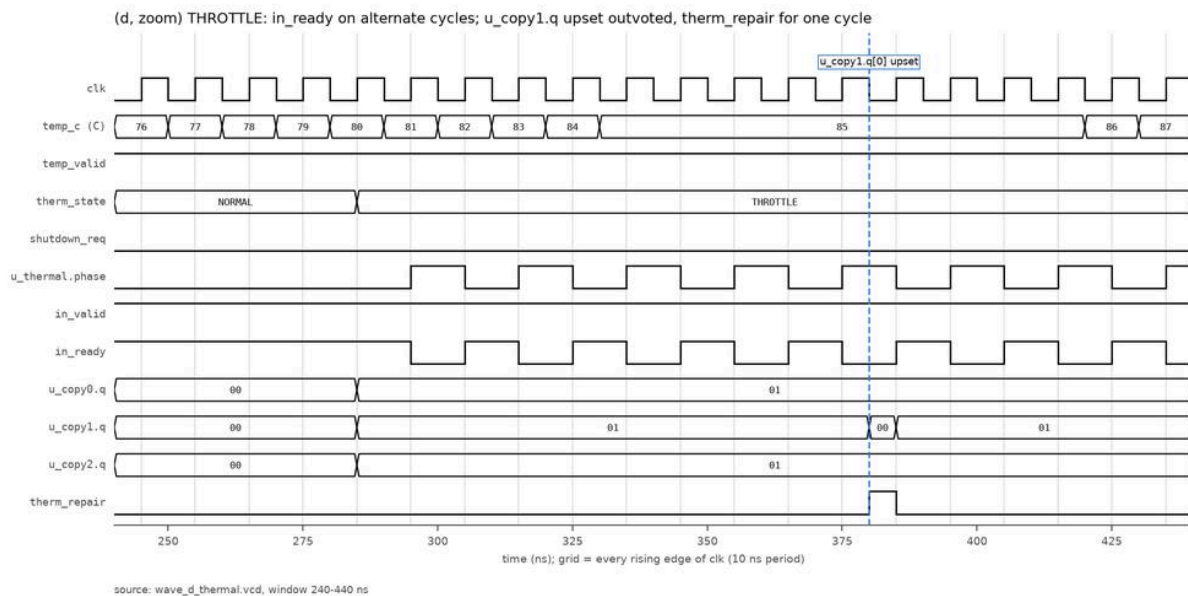


Figure 6, scenario 4, window 240-440 ns. NORMAL → THROTTLE at the 285 ns edge, after the 80 C reading; in_ready high on alternate cycles. u_thermal.u_copy1.q[0] is flipped at the 380 ns falling edge (copy1 01 → 00, copy0 = copy2 = 01); therm_repair = 1 until the 385 ns rising edge, the voted state stays THROTTLE, the admission pattern is unchanged and copy1 is rewritten from the voted next state at that edge. The log prints this event at t=381.0 ns, 1 ns after the flip. Same self-check and sources as Figure 5.

5 PVT corners and timing

Item	Status	Detail
PVT corners for RTL simulation (S1-S5, S12)	N/A	Functional simulation has no process, voltage or temperature. temp_c is a digital input of the thermal FSM, not a simulation temperature.
PVT corners for zero-delay GLS (S6-S9)	N/A	S7-S9: cell models are the Boolean functions of the sky130_fd_sc_hd__tt_025C_1v80 liberty (Yosys read_liberty ... write_verilog, mk/pdsep.mk lines 68-70 and 130-131). S6: Yosys generic gates (\$_AND_, \$_SDFFE_PN0P_ ...), no library (mk/sim.mk lines 104-108). No delay is used, so no corner applies.
SDF-annotated (timing) gate-level simulation, any corner	MISSING	No .sdf exists under build/ (searched 2026-10-06) and no script in mk/, pd/ or scripts/ calls write_sdf. Needed: OpenSTA write_sdf per corner from 6_final.odt and 6_final.spef.gz, sky130_fd_sc_hd Verilog models with specify blocks, an SDF-capable simulator flow, and the bench clock set to 7.2 ns.
Multi-corner timing	REPRODUCED 2026-10-06	Reported in 07_verification: multi-corner STA of the sep layout at 7.2 ns, sta_corners_sep.txt (TT, ss_100C_1v60, ff_n40C_1v95). ss_100C_1v60 fails setup there (WNS -6.131 ns). No simulation in this section covers or contradicts that result.
Simulation switching activity for power	MISSING	Power in 07_verification uses OpenSTA default activity (power_default_activity.txt). No SAIF/VCD from a gate-level workload exists; the waveform VCDs are short RTL scenarios and are not usable as power activity. Needed: a SAIF/VCD dump from a representative gate-level run of the sep netlist, annotated in report_power.

6 Negative controls (run_negctl.sh)

[run_negctl.sh](#) copies rtl/*.v to a scratch directory per mutant, applies one sed edit, stops if the edit did not apply (cmp), and runs all 5 scenarios. rtl/ is only read. Result: [negctl_summary.txt](#), REPRODUCED 2026-10-06 (Icarus 14.0 (devel), same conditions as section 4). The per-mutant copies and diffs are in the scratch wave_negctl/ directory, not in the package.

Mutant	File and change	Target scenario	Target result	Other scenarios
zext_product	orbit_mac_lane.v: prod32 = {16'd0, prod} (product zero-extended)	1 a_mac3	FAIL, 3 of 29 checks	b FAIL 2/35, c FAIL 1/42, c2 FAIL 1/50, d PASS
no_backpressure	orbit_demo.v: the buffer term of in_ready (~out_valid_q or out_ready) replaced by 1'b1	2 b_backpressure	FAIL, 2 of 35	all others PASS
out_valid_ungated	orbit_demo.v: out_valid = out_valid_q & ~fault_q (no ~mismatch)	3 c_fault	FAIL, 3 of 42	c2 FAIL 3/50, others PASS
fault_not_sticky	orbit_demo.v: fault_q <= 1'b1; else fault_q <= 1'b0; (fault follows mismatch)	5 c2_fault_sticky	FAIL, 3 of 50	all others PASS, including c
throttle_every_cycle	orbit_thermal_tmr.v: admit (voted == S_THROTTLE) without & ~phase	4 d_thermal	FAIL, 12 of 383	all others PASS
copy1_no_repair	orbit_thermal_tmr.v: u_copy1 .d(nxt) → .d(c1) (copy1 never rewritten)	4 d_thermal	FAIL, 92 of 383	a FAIL 1/29, others PASS

fault_not_sticky behaves like mutation-campaign survivor m22 (fault_q <= mismatch; in build/mutation/m22/rtl/orbit_demo.v), which passes sim, formal-quick, fault-quick, synth and the full formal set ([summary.md](#)). Scenario c2 kills it, but c2 is not part of any make target (open item 4).

Other negative controls for this section: S10 (26 sim mutants, 26/26 per bench, VERIFIED) and S11 (mutation campaign). For the S9 harness, a netlist mutant (instance _4098_sky130_fd_sc_hd__and2_1 → or2_1, 3000 cycles, seed 1, first MISMATCH at cycle 7, "GLS FAIL: 1148 errors") exists for the BASE netlist (reports/pd/sky130hd/gls_mutant.log, "#0 of 189"; the published file is cut at 8 lines and omits the verdict, which is in the full log build/pd/gls/sky130hd/mutant/gls_mutant.log, register D-49) and for the sep netlist ([gls_mutant.log](#), "#0 of 184", REPRODUCED 2026-10-06). The logs do not name their netlist; apart from that count they are identical.

7 Re-run comparison (committed 2026-09-30 vs re-run 2026-10-06)

Item	Committed	Re-run 2026-10-06	Difference
Directed, Icarus	directed_icarus.log	directed_icarus.log	none (byte-identical, md5 05a30e4e...)

Item	Committed	Re-run 2026-10-06	Difference
Directed, Verilator	directed_verilator.log lines 30-31: walltime 0.734 s, 1.819 ms/s, cpu 0.733 s, 8 MB	directed_verilator.log : walltime 1.326 s, 1.008 ms/s, cpu 1.301 s, 7 MB	runtime footer only; summary block identical. The default re-run reused the existing Verilator objects (verilator_build.log : "Nothing to be done", "Built from 0.000 MB sources in 0 modules"). A clean build in a scratch BUILD (verilator_build_clean.log : "Built from 0.298 MB sources in 7 modules") also passed, walltime 1.253 s, 8 MB, "summary identical to Icarus" (make_sim-verilator_cleanbuild.log)
Model self-test	model.log	model.log	none
cocotb, Icarus	random_icarus.md : wall s 2.4 / 2.7 / 2.7 / 2.4 / 2.5 / 19.0	random_icarus_summary.md : 3.6 / 3.6 / 3.3 / 3.2 / 3.3 / 24.9	"wall s" column only; cycles, results, errors and the full coverage table identical. make_sim-random-verilator re-ran the Icarus step once more (3.2 / 3.5 / 3.3 / 3.3 / 3.3 / 24.6 s, same counts)
cocotb, Verilator	random_verilator.md : wall s 1.6 / 1.5	random_verilator_summary.md : 2.1 / 2.1	"wall s" only. The cocotb Verilator build of 2026-09-30 was reused (build/sim/random_verilator/build.log: "Nothing to be done"); not rebuilt from scratch
Yosys generic GLS	gl_s_yosys.log	gl_s_yosys.log	none; 4899 cells / 521 flip-flop cells as summary.md line 58
Routed-netlist GLS	gl_sky130hd.log (BASE netlist)	gl_sky130hd_sep_directed.log (SEP netlist)	byte-identical logs from different netlists; the bench output does not depend on the netlist when the function matches. Not a re-run of the same input
Sim mutants (S10)	mutants.md	make_mutants.log lines 4-37	none (table byte-identical)
GLS fault campaign (S9)	gl_s.log	gl_s_rerun.log	none: lines 1-6 identical; real 2m17.734s

Re-run wall times (real, from the make_*.log files; no committed timings exist): lint 0.123 s, sim-model 0.045 s, sim-directed 6.527 s, sim-verilator 8.782 s cached / 28.536 s clean, sim-random 42.417 s, sim-random-verilator 46.156 s (includes the Icarus step), sim-gls Yosys 46.567 s, directed GLS on the sep netlist 2 min 3.381 s. The vacuity re-run log [make_formal-vacuity.log](#) in this folder belongs to 07_verification. Every functional output of the simulation re-runs with a committed counterpart (S1-S6, S9, S10) equals the committed one; the differences are limited to wall time, CPU time and memory fields. S8 and the lint run (S14) have no committed counterpart. In make_formal-vacuity.log all 19 mutants are CAUGHT, as in the committed [vacuity.md](#), but zext_product_pdr reports a different failing property and step (P8_fresh_sum_from_zero_lane2, step 6; committed P1_out_valid_data_lane1 P8_fresh_sum_from_zero_lane1, step 4) and clear_keeps_acc_pdr a different step (8; committed 5): abc_pdr run-to-run variation, register D-43.

8 Open items

#	Item	Status	Needed
1	Directed-bench GLS of the reviewed sep netlist exists only as this package's re-run (S8); the repo publishes the base netlist (S7) and tb/sim_report.py hardcodes the base path	MISSING	commit the S8 log; take the netlist path from SIM_GLS_SRCS. See discrepancy register (09_review_notes)
2	Timing (SDF) gate-level simulation at any corner	MISSING	see section 5
3	Fault path in the regression suite: the sim benches never inject upsets (fault, therm_repair checked = 0 only); thermal code 3 unreachable; mutation survivors m07, m20, m22, m27	MISSING	add the checks of tb_mutation_gaps.v to a make target
4	Waveform scenarios c, c2 and d are not in any make target. c2 kills a non-sticky fault latch (section 6), which the regression suite misses (m22)	MISSING	add waveforms/tb_orbit_wave.v +scen=1..5 to make_sim or move its checks into tb/
5	cocotb random test on a gate-level netlist	MISSING	run tb/cocotb/run_random.py --sources <sep_6_final.v> <cells.v>

#	Item	Status	Needed
6	Mid-run reset coverage in the random test (seed 2 has none; not a required point)	MISSING	add reset coverage points (during out_fire, in THROTTLE, in STOP) to REQUIRED
7	Code / toggle coverage	MISSING	e.g. Verilator --coverage on S1 and S3 with a merged report
8	Switching activity for power	MISSING	see section 5
9	Provenance in the sim logs (commit, RTL/TB hash, tool versions; summary.md says "unknown")	MISSING	have tb/sim_report.py write git rev-parse HEAD, sha256 of \$(RTL) and the bench, and tool versions. See discrepancy register (09_review_notes)
10	Re-run evidence formerly not in the package: sim-mutants re-run (S10)	resolved in package: make_mutants.log (REPRODUCED 2026-10-06)	—
11	From-scratch build of the cocotb Verilator model	MISSING	re-run make sim-random-verilator with an empty BUILD
12	cocotb is not listed in requirements-tools.txt (pinned only indirectly through oss-cad-suite==2026-09-28, line 6); Python is pinned only as python==3.11 (line 16), and the python3 used by make sim-model is not logged. Line 15 says the scripts use the standard library only, although the cocotb bench needs cocotb (register D-15)	MISSING	add cocotb to requirements-tools.txt; print python3 --version in the model log
13	No demonstrator clock requirement: docs/SPEC.md states no frequency; all simulations use 10 ns, the layout 7.2 ns	MISSING	state the target clock and PVT range in SPEC
14	Waveform bench labels: run_scen4.log uses "cyc" for the bench cycle counter in event lines (e.g. cyc=37) and for the ramp step in the final line (first THROTTLE cyc 24); the usage comment in tb_orbit_wave.v line 34 reads "+scen=1 (2, 3, 4)", omitting scenario 5 (c2_fault_sticky) that the header (lines 10-23) and run_waves.sh run	MISSING (labelling fix not applied; no result affected)	rename the final-line field to "step"; change the usage comment to "+scen=1 (2, 3, 4, 5)"